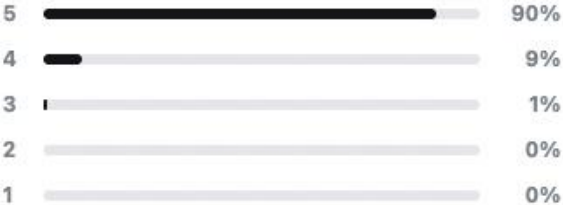

Aspect-Based Sentiment Analysis on Customer Review

Naiska Buyandalai

Global Reviews (999+)

Korean Reviews (999+)

4.9 
9,627 reviews



Sun protection 
Texture 

Write a Review

Photo & Video (999+)

View more >



9,627 reviews

Most Helpful



 Best

SKIN1004 Madagascar Centella Hyalu-Cica Water-Fit Sun Serum 50mL*2ea (Twin pack)

Texture  Sun protection 

This is a recurring purchase for me. I first bought this sunblock based on the recommendation of a sales associate at Olive Young COEX last winter. After finishing another Korean sunblock, I brought it back to Singapore and started using it. In one word: I love this sunblock! It is very lightweight, and the quality is outstanding. There is no scent, and it glid...

[Translate](#) | [Read more](#)



 5

yy*****
2025/07/28

Skin Tone Medium

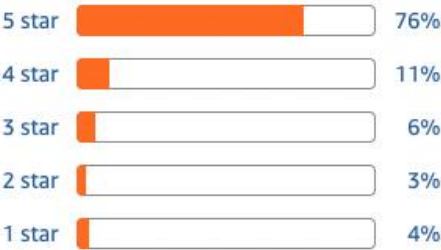
Skin Type Oily

Skin Concern Acne ·
Brightening ·
Dullness

Customer reviews

★★★★☆ 4.5 out of 5

862 global ratings



How customer reviews and ratings work

Review this product

Share your thoughts with other customers

Write a customer review

Customers say

Customers find this moisturizer effective at hydrating dry skin and appreciate its gentle, non-greasy formula. The product receives positive feedback for its lightweight texture, and one customer notes it leaves skin looking fresh and clear without a white cast. Regarding skin irritation, while many report no issues, some customers mention it can be irritating.

Generated from the text of customer reviews

Select to learn more

- ✓ Hydration (34)
- ✓ Moisturizer quality (29)
- ✓ Lightweight (18)
- ✓ Skin softness (18)
- ✓ Oiliness (17)
- ✓ Scent (10)
- ✓ Skin clarity (10)
- Skin irritation (13)

Reviews with images

See all photos



Top reviews from the United States



★★★★★ **Miracle Lotion**

Reviewed in the United States on October 24, 2025

Verified Purchase

This is the best lotion by far. It just takes away any sort of dry patches and makes the skin look so smooth and perfect. I am so happy I discovered this through a YouTuber. Holy Grail product.

2 people found this helpful

Helpful | Report

Why Aspect-Based Sentiment Analysis?

Traditional Sentiment Analysis

- One sentiment for the entire review
- Misses mixed opinions in the same text
- Good for simple "positive/negative" summaries
- Cannot tell why a review is positive or negative

Aspect-Based Sentiment Analysis (ABSA)

- Sentiment per aspect within a review
- Good for customer insights
- Identifies specific strengths and weaknesses
- Treats each aspect-sentiment pair as one prediction

Dataset Overview

FABSA Dataset (from HuggingFace)

Total Reviews:	Train: 7,930
	Validation: 1,057
	Test: 1,587
Total Flattened Aspect Level Dataset:	Train: 13,998
	Validation: 1,858
	Test: 2,812

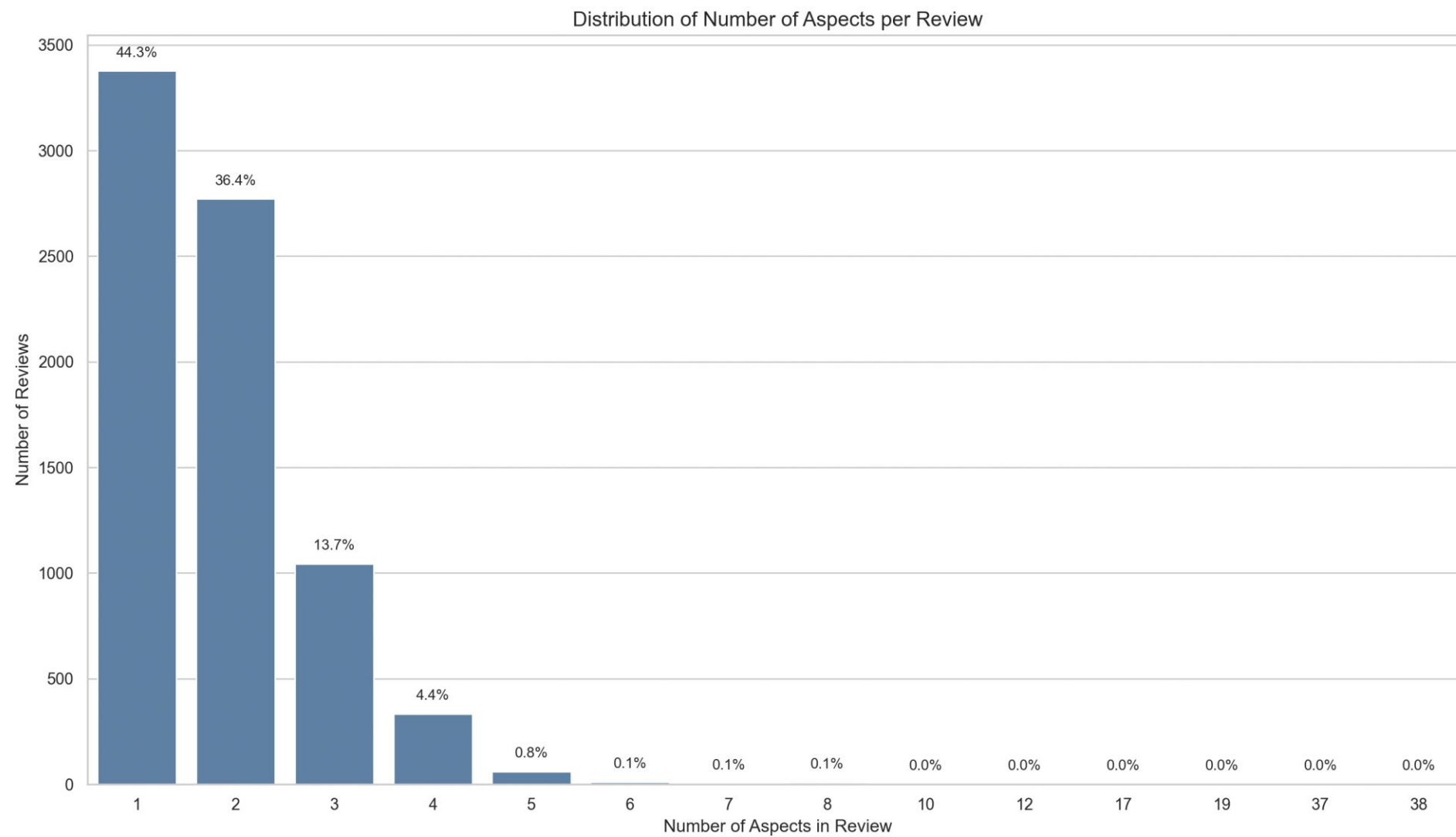
Dataset Overview

Column Name	Description
id	Unique identifier for each review in the dataset
data_source	Platform where the review was collected (Trustpilot, Google Play, App Store)
org_id	Organization id
industry	Category of the business being reviewed (Banking, Fashion, Groceries, Travel etc.)
text	Full review text written by the user
labels	List of (aspect, sentiment) pairs extracted from the review. Example: <code>[("Ease of use", "positive"), ("Customer support", "negative")]</code>
label_codes	Numerical codes for each aspect sentiment pair

```
id: 301972057
org_index: 600
data_source: Trustpilot
industry: Price Comparison
text: My experience is only around the Parking forum, so my review is based on this specific experience. As someone who needed...
labels: [['Staff support: Attitude of staff', 'negative'], ['Company brand: Reviews', 'negative'], ['Company brand: General sati...
label_codes: ['staff-support.attitude-of-staff.-1', 'company-brand.reviews.-1', 'company-brand.general-satisfaction.-1']
```

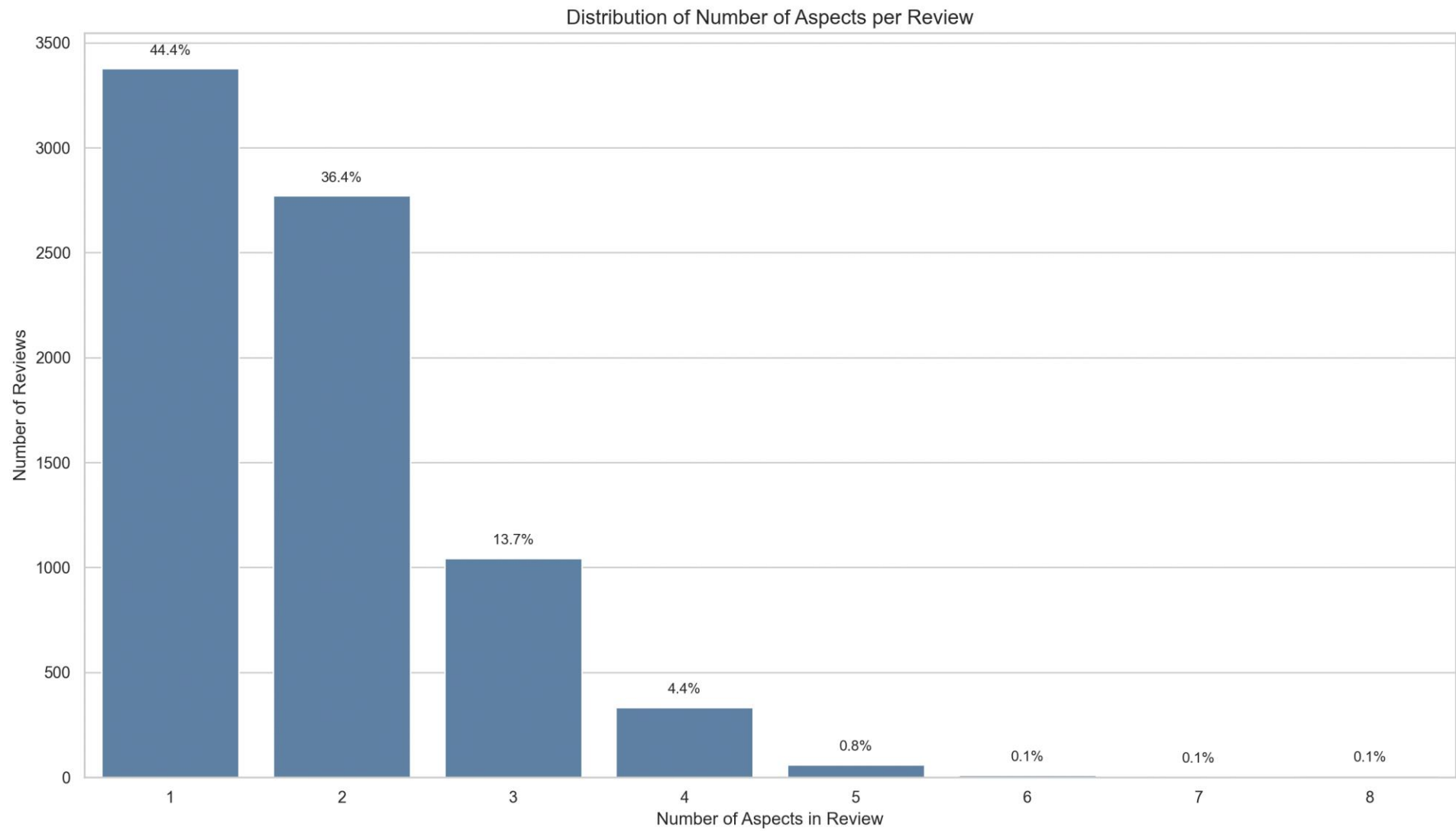
Dataset preprocessing

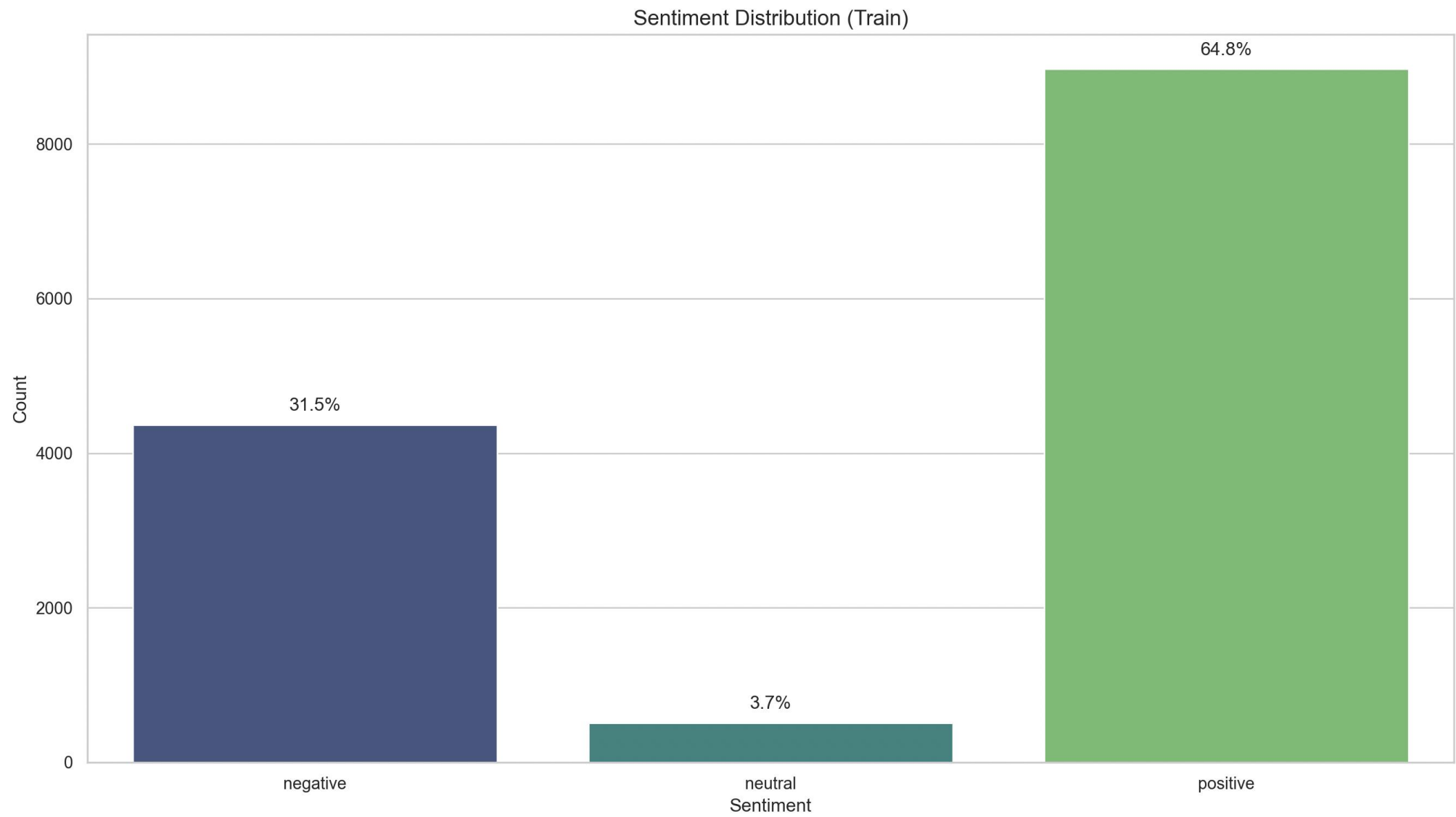
1. Flattened the FABSA dataset from review-level to aspect-level
 2. Removed a small number of outlier reviews with more than 8 aspects
 3. Removed the anonymized placeholder "org"
 4. Lowercased text, removed punctuation and non-letter characters, and collapsed extra spaces.
 5. Tokenized text and removed stopwords
-

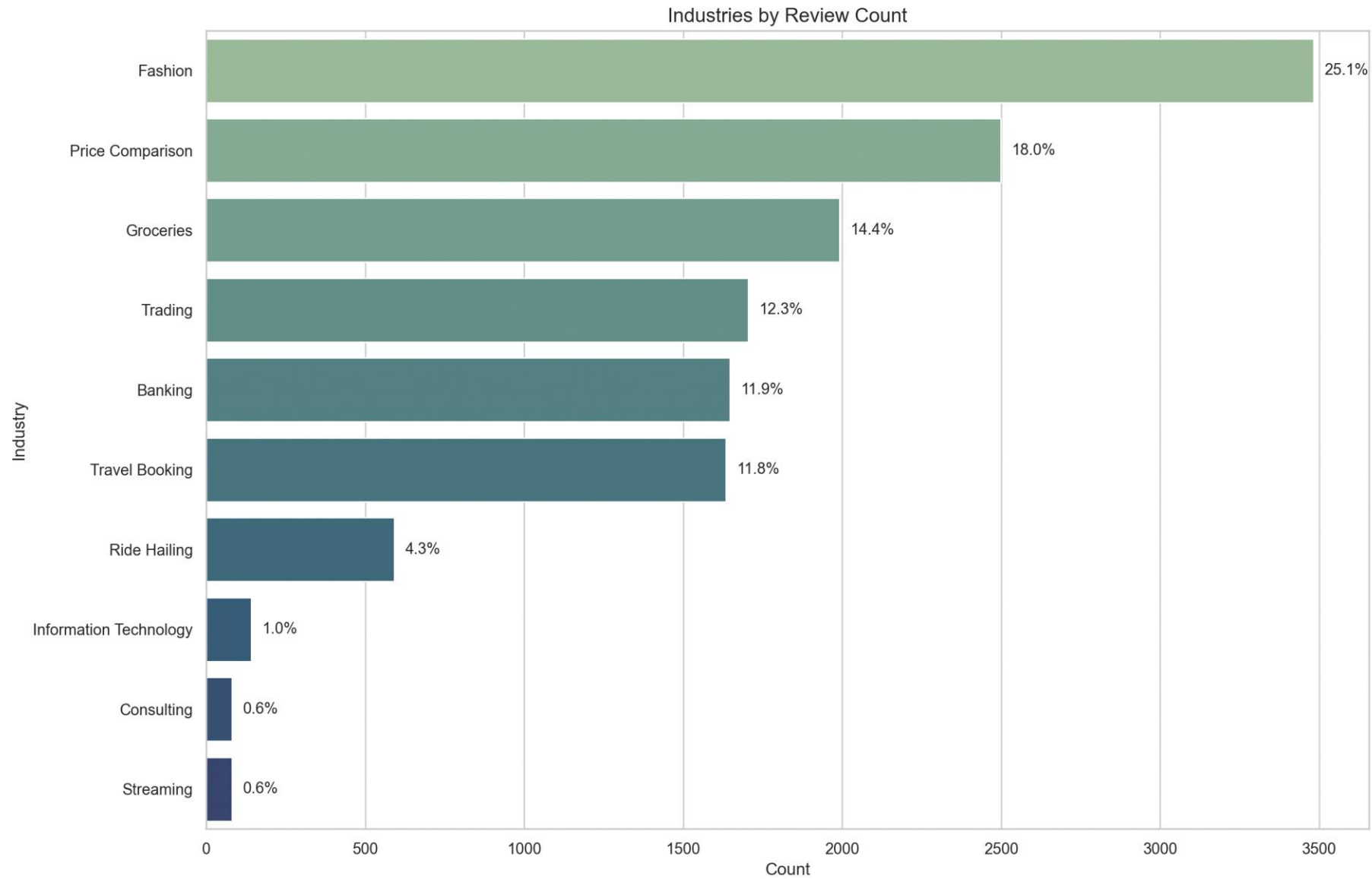


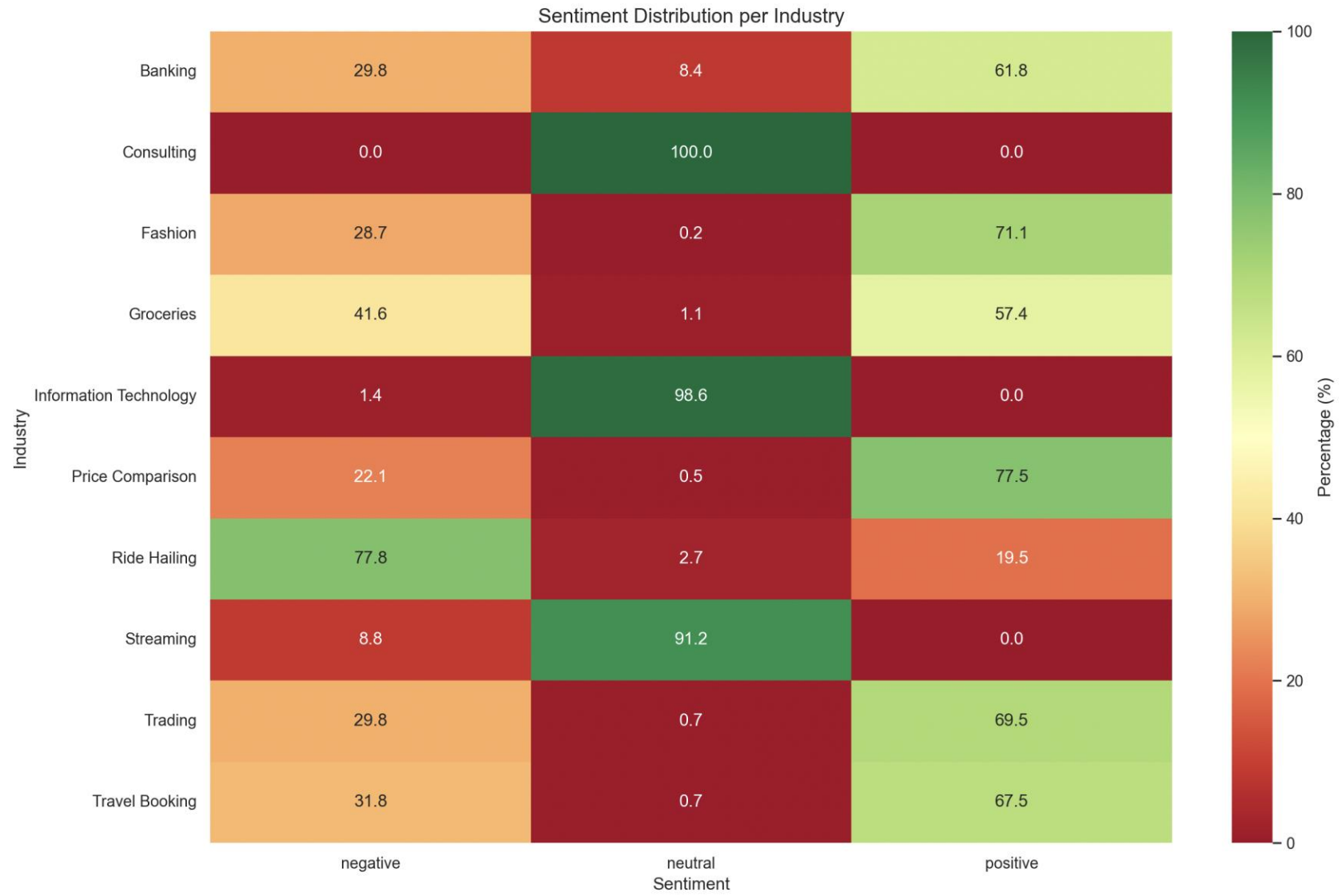


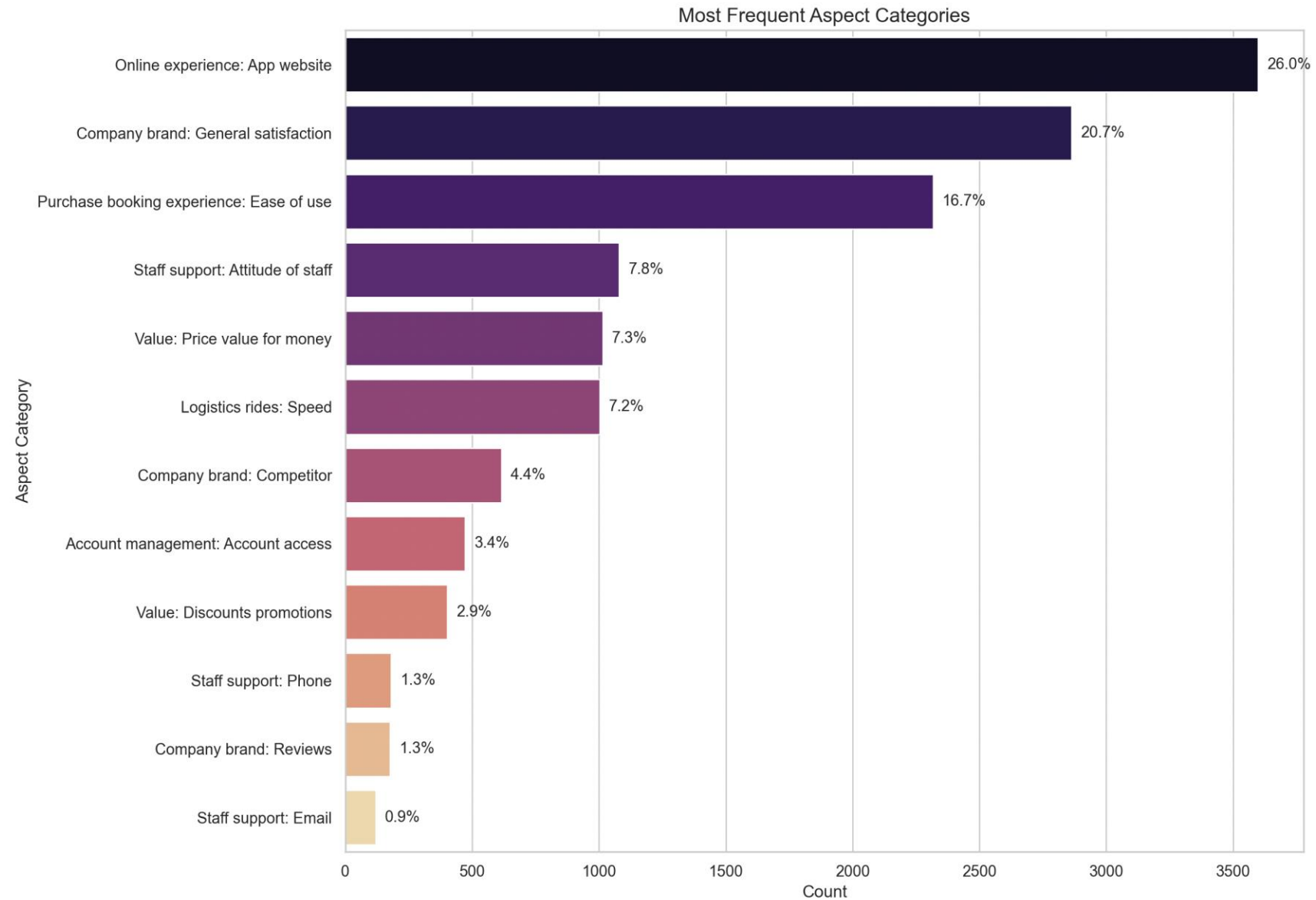
Analysis

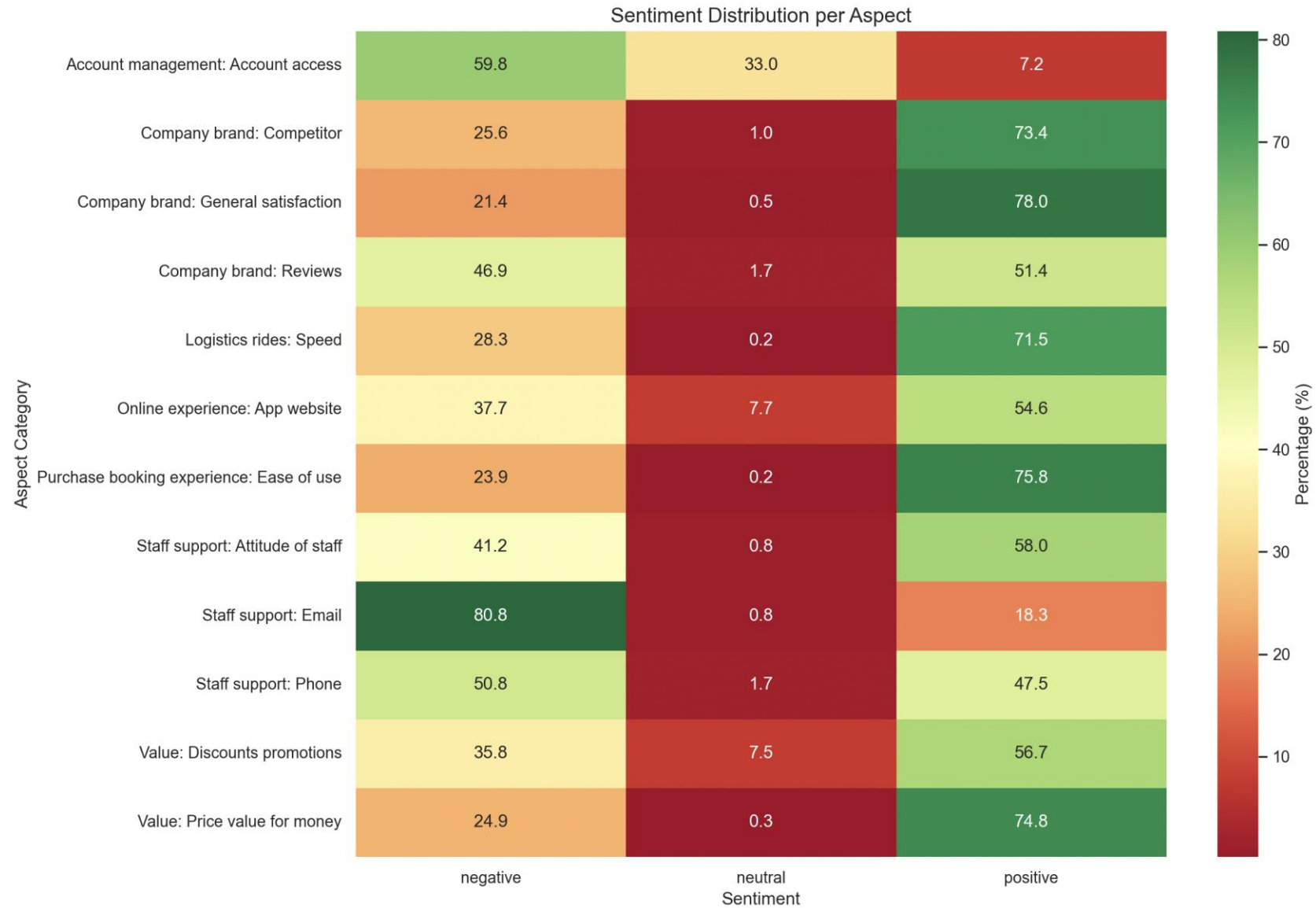




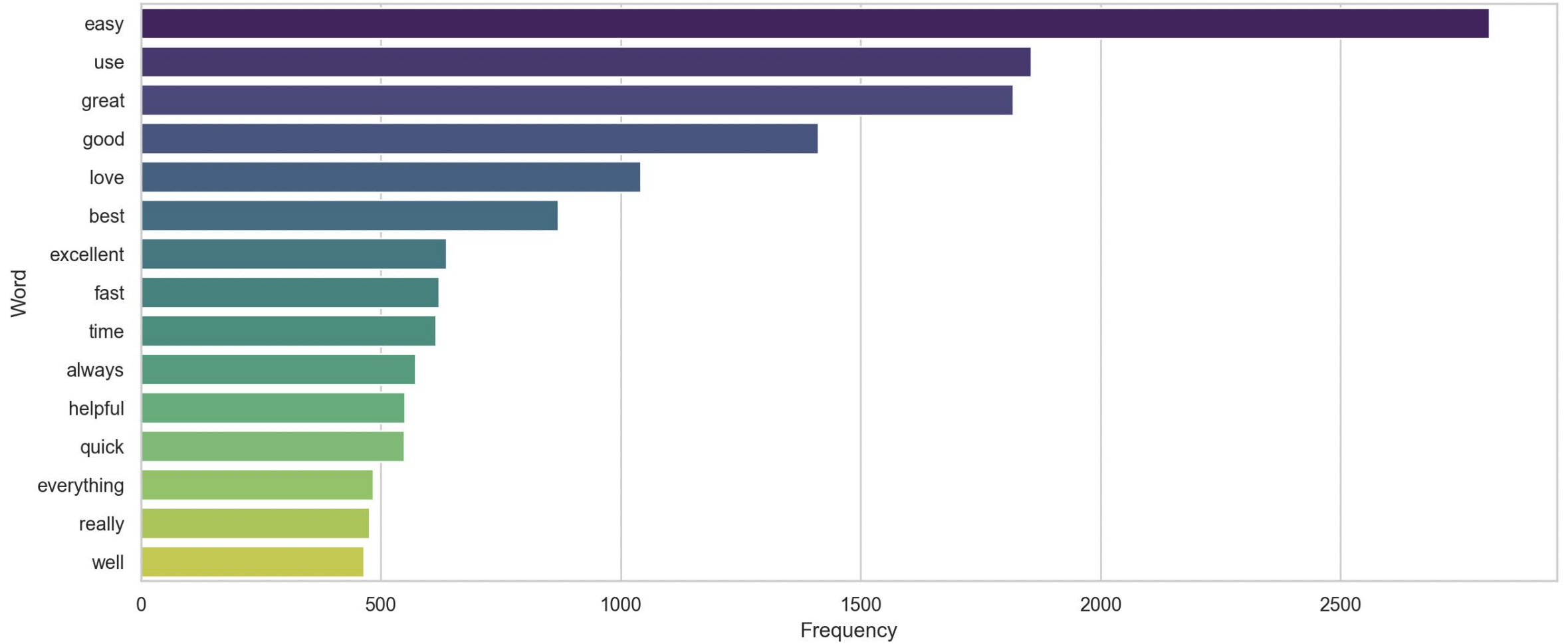




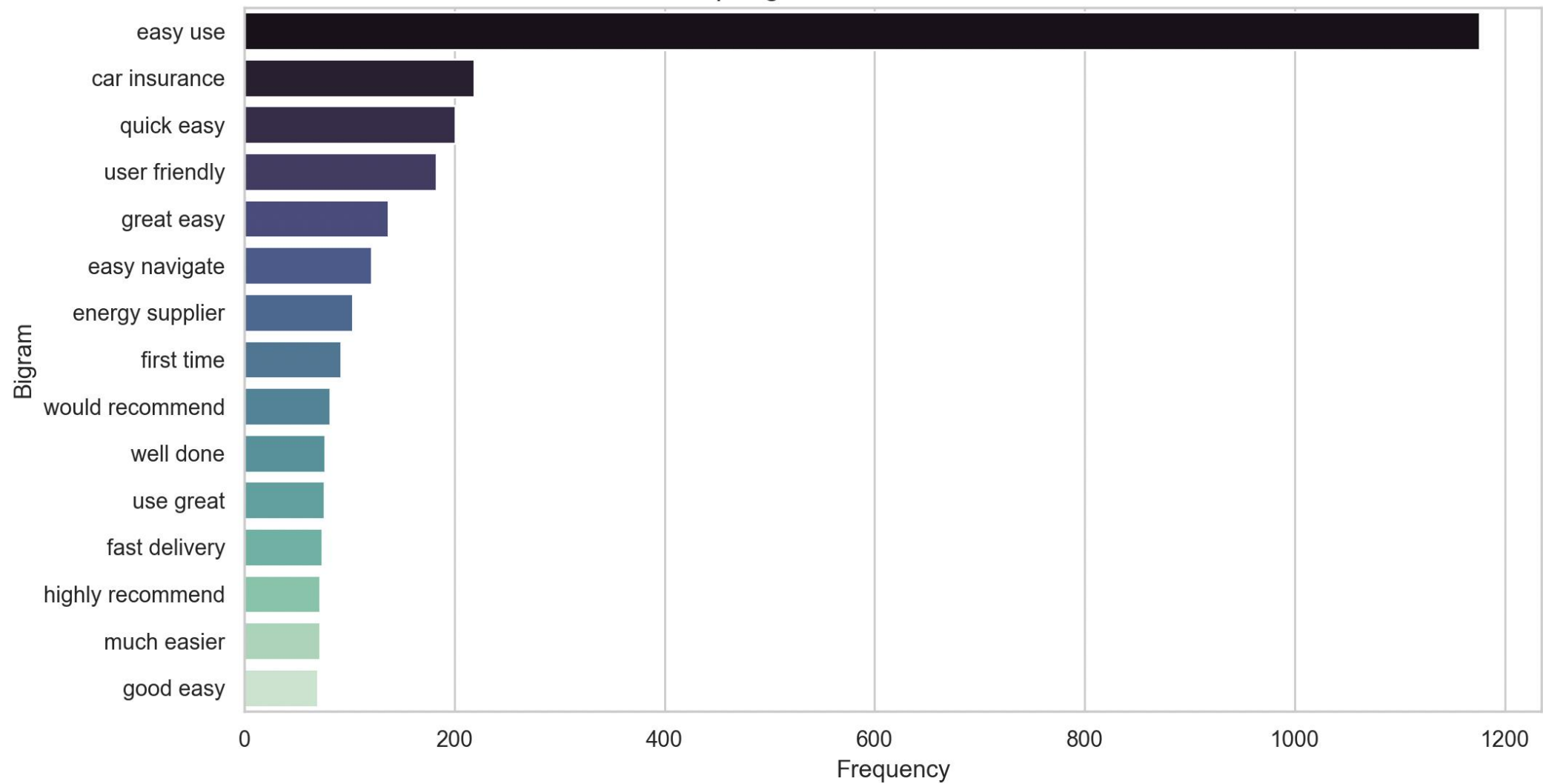




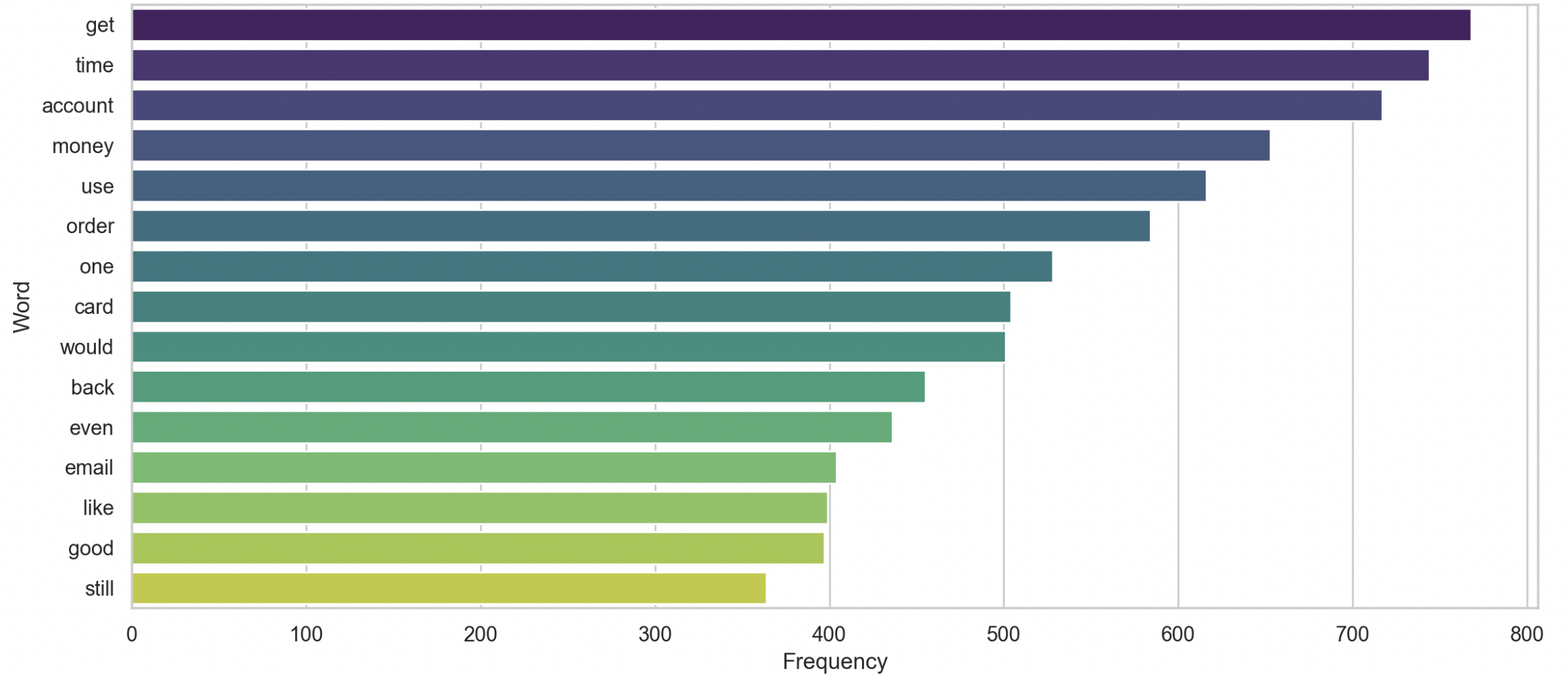
Top Words — Positive Reviews



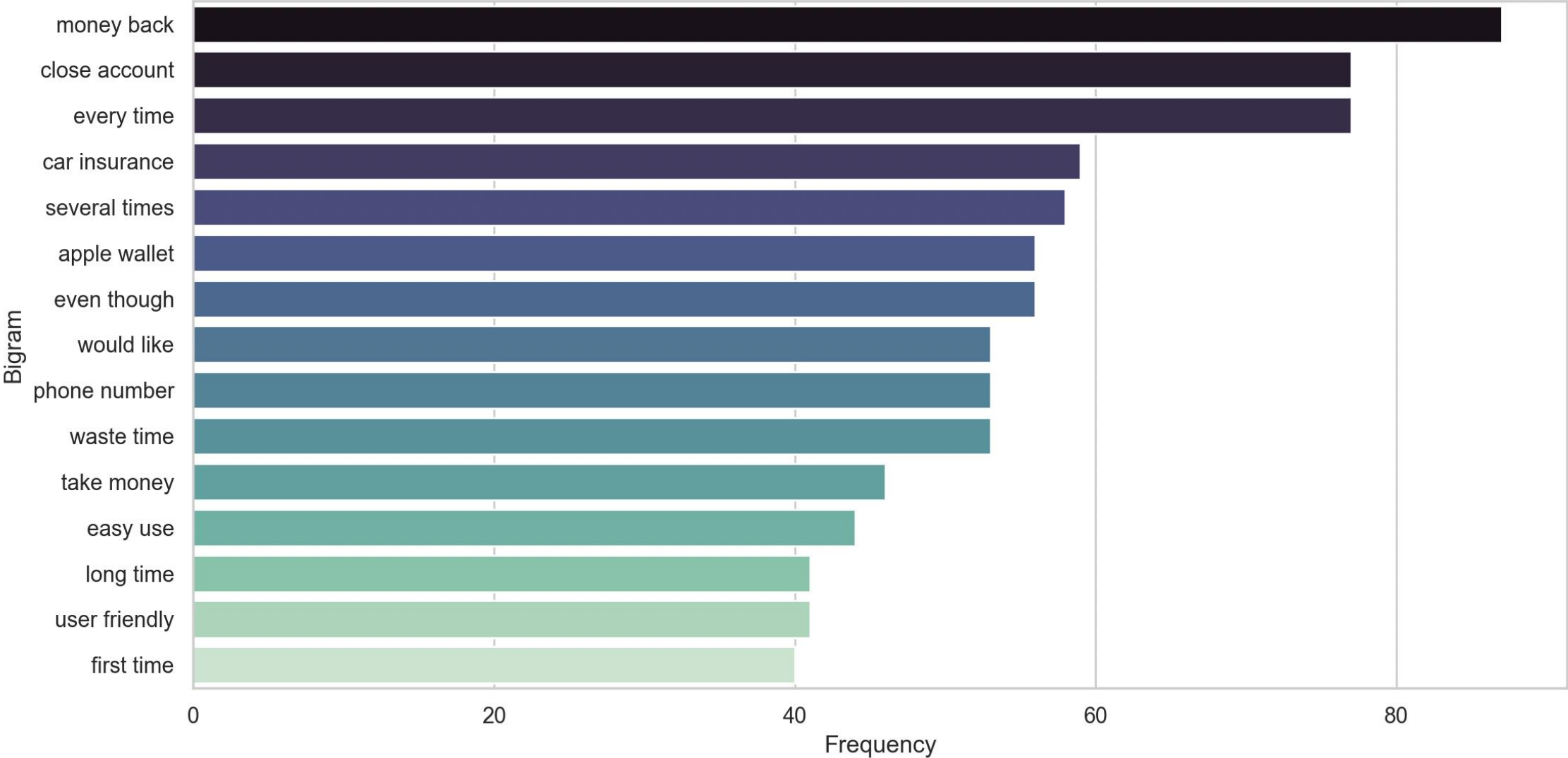
Top Bigrams — Positive Reviews



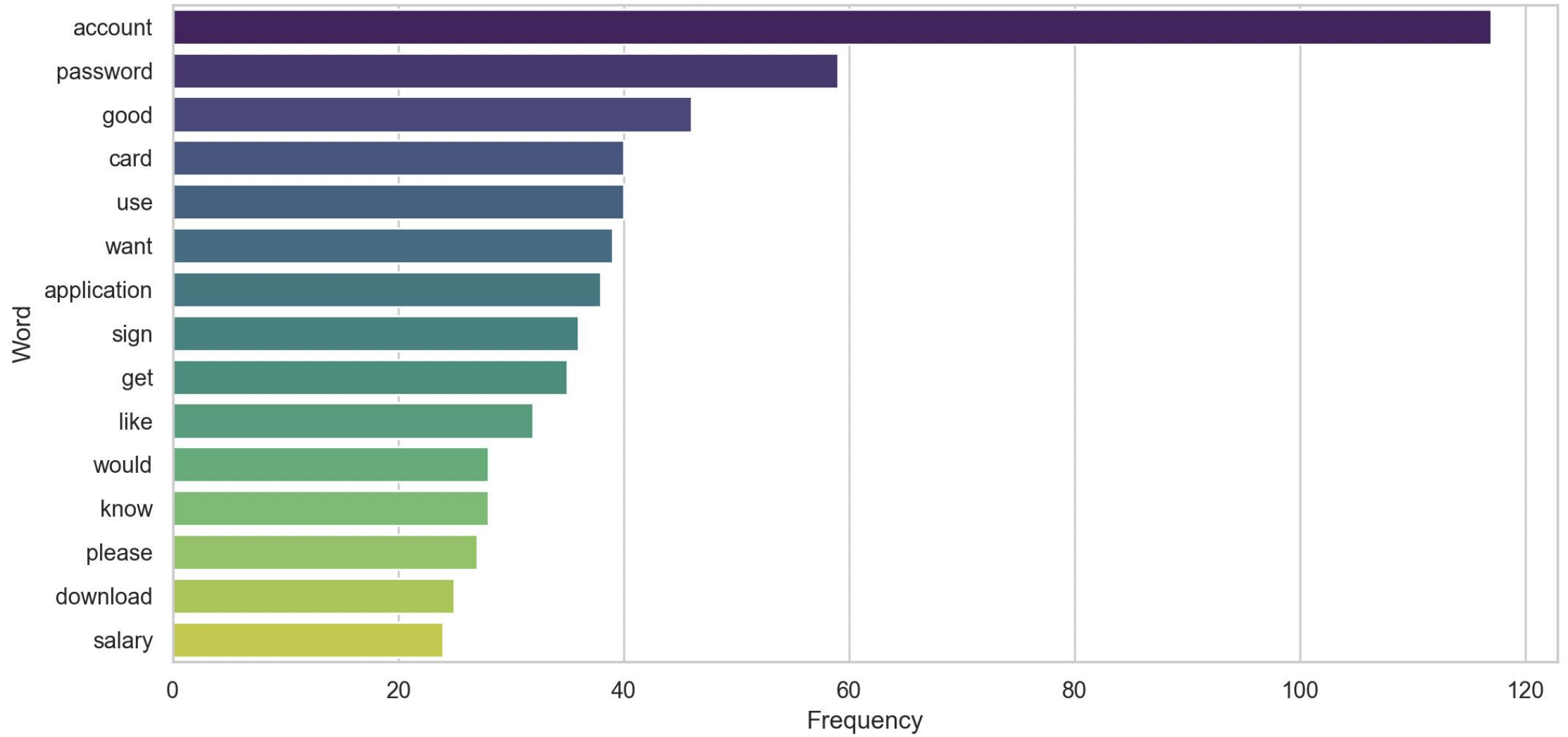
Top Words — Negative Reviews



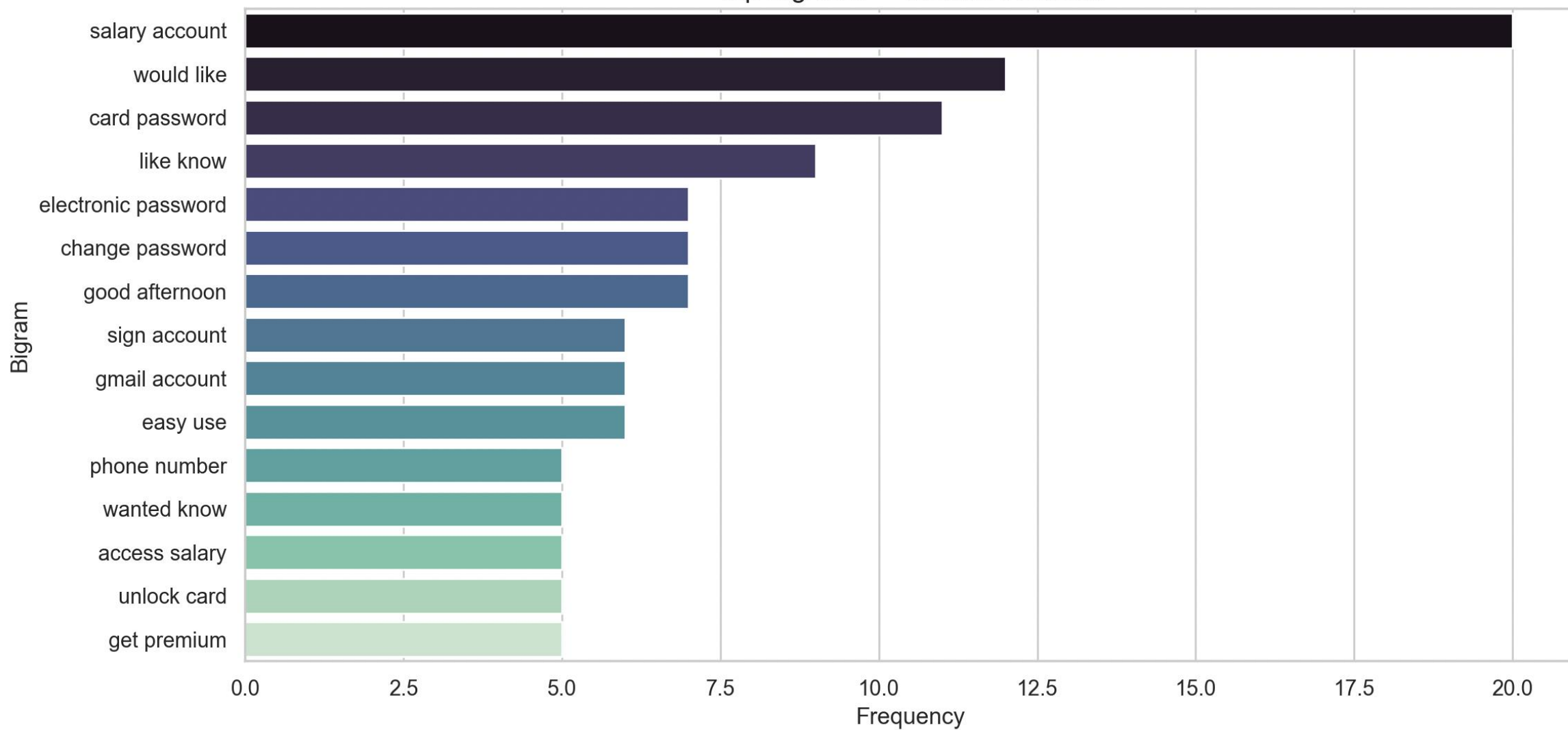
Top Bigrams — Negative Reviews

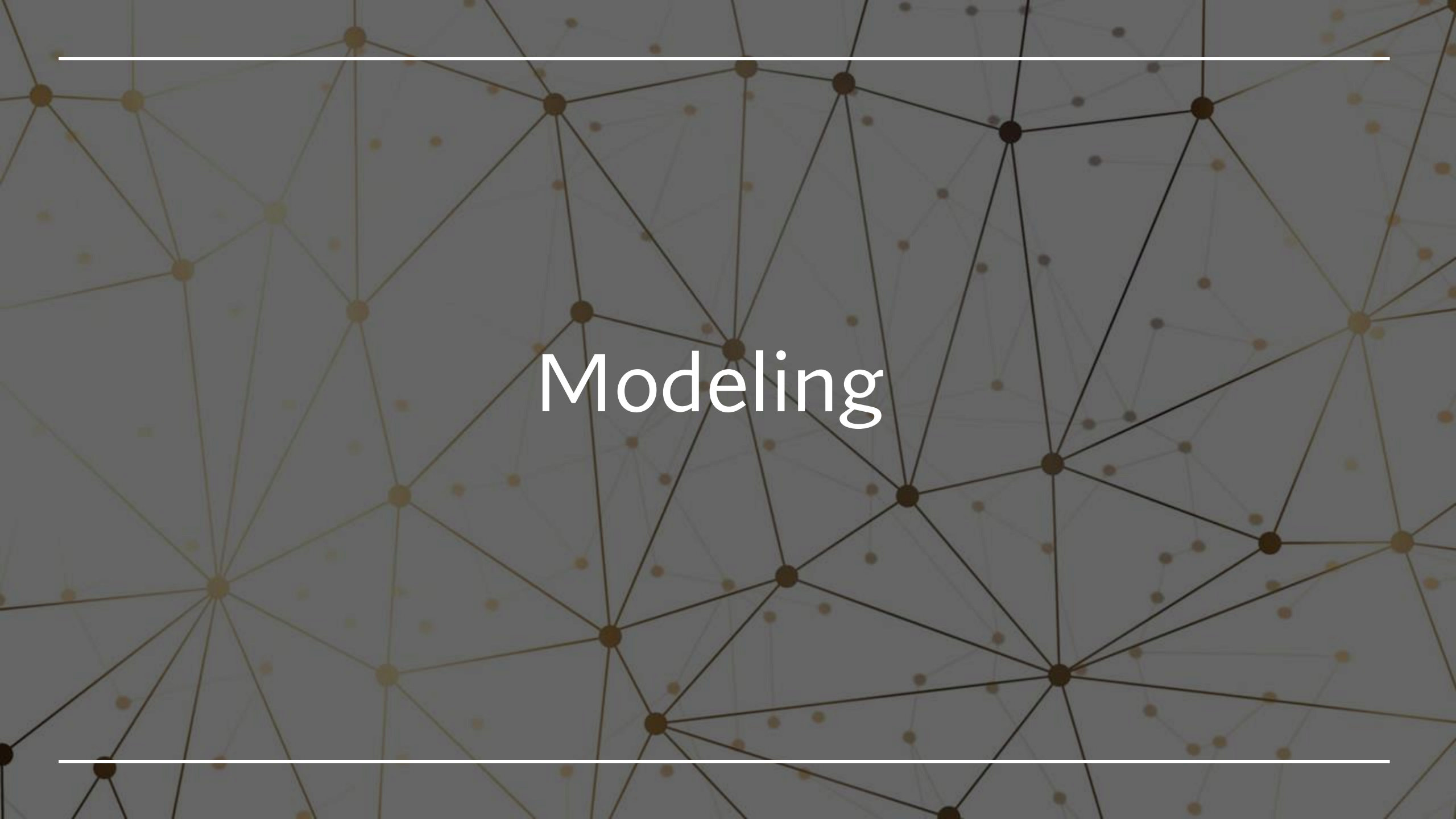


Top Words — Neutral Reviews



Top Bigrams — Neutral Reviews





Modeling

Modeling Approach

Input: customer review text, aspect category

Output: sentiment (negative, neutral, positive)

Model: roBERTa-base

Tokenizer: roBERTa BPE

Training

Optimizer: **AdamW**

Learning rate: **$1e-5$**

Batch size: **16**

Epochs: **5**

Max sequence length: **128 tokens**

Handling Class Imbalance

class-weighted CrossEntropyLoss

- Weight negative ≈ 3.21
- Weight neutral ≈ 27.45
- Weight positive ≈ 1.53

```
Classification report:
              precision    recall  f1-score   support

   negative      0.90      0.88      0.89       896
    neutral      0.86      0.79      0.82        91
   positive      0.94      0.96      0.95      1825

 accuracy              0.93       2812
macro avg      0.90      0.88      0.89       2812
weighted avg      0.93      0.93      0.93       2812
```

```
Classification report:
              precision    recall  f1-score   support

   negative      0.84      0.95      0.89       896
    neutral      0.67      0.85      0.75        91
   positive      0.98      0.91      0.94      1825

 accuracy              0.92       2812
macro avg      0.83      0.90      0.86       2812
weighted avg      0.93      0.92      0.92       2812
```

The background features a complex network of nodes and lines. Nodes are represented by small circles in various colors including brown, gold, and grey. They are interconnected by thin, dark lines, creating a web-like structure. The overall color palette is muted, with a dark grey background and earthy tones for the nodes and lines.

SHAP

Sentiment: Negative

Aspect : Online experience: App website

Industry: Ride Hailing

Source : Google Play

“easy to use.gud response from customer care sections. peak tym price hike n morning 9-10.30 rates should be verified..sometimes the app showing double price for ORG62 services. 10-15% price hike can be considerable .your current -utilizing the needy ppl - is not a gud business approach .thank you ORG62 team.pls review my suggestion”

Staff support: Attitude of staff - positive

Online experience: App website – negative

Purchase booking experience: Ease of use – positive

Value: Price value for money - negative

```
Model probabilities for [negative, neutral, positive]:
Sample 0 true=negative probs=[0.8585763  0.00250881 0.13891494]
Sample 1 true=neutral probs=[1.7646494e-03 9.9785405e-01 3.8122205e-04]
Sample 2 true=positive probs=[1.2791331e-03 3.7091161e-04 9.9834991e-01]
```

Negative

```
'easy ' shap=-0.2913
'thank ' shap=-0.1401
'not ' shap=+0.1112
'double ' shap=+0.0699
'price ' shap=+0.0502
'you ' shap=-0.0444
'rates ' shap=+0.0422
'sections. ' shap=+0.0387
'needy ' shap=+0.0374
'should ' shap=+0.0363
```

Sample 0
Neutral

```
'pls ' shap=+0.0461
'team.' shap=-0.0305
'ORG62 ' shap=-0.0252
'app ' shap=+0.0236
'not ' shap=-0.0202
'easy ' shap=-0.0176
'considerable .' shap=-0.0164
'double ' shap=-0.0150
'App ' shap=+0.0138
'price ' shap=-0.0137
```

Positive

```
'easy ' shap=+0.3088
'thank ' shap=+0.1493
'not ' shap=-0.0911
'double ' shap=-0.0549
'you ' shap=+0.0420
'sections. ' shap=-0.0377
'sometimes ' shap=-0.0372
'price ' shap=-0.0365
'rates ' shap=-0.0326
'should ' shap=-0.0314
```

Sentiment: Neutral

Aspect : Account management: Account access

Industry: Banking

Source : Google Play

“How do I access my card?”

Account management: Account access - neutral

```
Model probabilities for [negative, neutral, positive]:
Sample 0 true=negative probs=[0.8585763  0.00250881 0.13891494]
Sample 1 true=neutral probs=[1.7646494e-03 9.9785405e-01 3.8122205e-04]
Sample 2 true=positive probs=[1.2791331e-03 3.7091161e-04 9.9834991e-01]
```

Negative

```
'Account ' shap=-0.1986
'Account ' shap=-0.1834
'access ' shap=-0.1552
'I ' shap=-0.1342
'management: ' shap=+0.0839
'card' shap=+0.0690
'do ' shap=-0.0669
'How ' shap=-0.0501
'access ' shap=-0.0451
'my ' shap=-0.0255
```

Sample 1 Neutral

```
'Account ' shap=+0.2279
'Account ' shap=+0.2198
'access ' shap=+0.1781
'I ' shap=+0.1457
'do ' shap=+0.0751
'management: ' shap=-0.0722
'How ' shap=+0.0706
'access ' shap=+0.0656
'card' shap=-0.0077
'my ' shap=-0.0002
```

Positive

```
'card' shap=-0.0613
'Account ' shap=-0.0364
'Account ' shap=-0.0293
'my ' shap=+0.0257
'access ' shap=-0.0229
'access ' shap=-0.0205
'How ' shap=-0.0204
'management: ' shap=-0.0117
'I ' shap=-0.0115
'do ' shap=-0.0082
```

Sentiment: Positive

Aspect : Purchase booking experience: Ease of use

Industry: Banking

Source : Google Play

“Very useful and easy.”

Purchase booking experience: Ease of use - positive

```
Model probabilities for [negative, neutral, positive]:
Sample 0 true=negative probs=[0.8585763  0.00250881 0.13891494]
Sample 1 true=neutral probs=[1.7646494e-03 9.9785405e-01 3.8122205e-04]
Sample 2 true=positive probs=[1.2791331e-03 3.7091161e-04 9.9834991e-01]
```

Negative

Sample 2
Neutral

Positive

```
'easy'      shap=-0.3591
'useful'    shap=-0.2348
'Very'      shap=-0.1105
'Ease'      shap=-0.0907
'use'       shap=+0.0195
'Purchase'  shap=+0.0193
'experience:' shap=+0.0168
'of'        shap=+0.0163
'booking'   shap=+0.0121
'and'       shap=+0.0044
```

```
'easy'      shap=-0.0387
'Ease'      shap=-0.0113
'booking'   shap=-0.0090
'and'       shap=-0.0087
'useful'    shap=-0.0080
'Very'      shap=-0.0067
'use'       shap=-0.0067
'of'        shap=-0.0051
'Purchase'  shap=-0.0020
'experience:' shap=+0.0014
```

```
'easy'      shap=+0.3978
'useful'    shap=+0.2428
'Very'      shap=+0.1172
'Ease'      shap=+0.1021
'experience:' shap=-0.0182
'Purchase'  shap=-0.0173
'use'       shap=-0.0128
'of'        shap=-0.0112
'and'       shap=+0.0044
'booking'   shap=-0.0031
```